

BMI500: HW3 - Profiling Exercise Report

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(a) Repeat the profiling exercise in the lab for pbmc6k and pbmc10k datasets (coarse-grain, instrumented, fine-grain).

(i) Pbm3k

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:03)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
```

Coarse_grain_profiling (without cached files - true cold start)

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:03)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
  Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc3k --out-dir out"
  User time (seconds): 23.46
  System time (seconds): 5.36
  Percent of CPU this job got: 222%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:12.93
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 870476
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 0
  Minor (reclaiming a frame) page faults: 298249
  Voluntary context switches: 19671
  Involuntary context switches: 446125
  Swaps: 0
  File system inputs: 0
  File system outputs: 106931
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

Use “time.time()” to measure elapsed time for each code section marked with “###” (instrumented profiling)

```
running Louvain clustering
using the "louvain" package of Traag (2017)
finished: found 4 clusters and added
'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.066s
computing UMAP
finished: added
'X_umap', UMAP coordinates (adata.obsm) (0:00:03)
[TIMING] umap: 3.508s
[TIMING] write results file: 0.151s
ranking genes
finished: added to '.uns['rank_genes_groups']'
'names', sorted np.recarray to be indexed by group ids
'scores', sorted np.recarray to be indexed by group ids
'logfoldchanges', sorted np.recarray to be indexed by group ids
'pvals', sorted np.recarray to be indexed by group ids
'pvals adj', sorted np.recarray to be indexed by group ids (0:00:01)
Tue Sep 15 22:43:23 2026 rank_genes_groups.prof

42485 function calls (41332 primitive calls) in 1.192 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>

ncalls  tottime  percall  cumtime  percall  filename:lineno(function)
24/1    0.000    0.000    1.192    1.192 {built-in method builtins.exec}
1       0.000    0.000    1.192    1.192 <string>:1(<module>)
1       0.000    0.000    1.192    1.192 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/legacy_api_wrap/_init_.py:83(fn_compatible)
1       0.000    0.000    1.192    1.192 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1       0.002    0.002    1.168    1.168 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
5       0.008    0.002    1.147    0.229 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
5       0.003    0.001    0.947    0.189 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
4       0.000    0.000    0.889    0.222 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10012(rank)
4       0.000    0.000    0.889    0.222 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
4       0.888    0.222    0.888    0.222 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
60/59   0.000    0.000    0.083    0.001 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1177(_getitem_)
41       0.000    0.000    0.081    0.002 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1690(_getitem_tuple)
57       0.000    0.000    0.081    0.001 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1720(_getitem_axis)
23       0.000    0.000    0.080    0.003 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1206(_getbool_axis)
23       0.000    0.000    0.079    0.003 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:4164(_take_with_is_copy)

[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 1.311s
```

(ii) Pbm6k

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
    'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
    'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
ranking genes
  finished: added to `uns['rank_genes_groups']`
    'names', sorted np.recarray to be indexed by group ids
    'scores', sorted np.recarray to be indexed by group ids
    'logfoldchanges', sorted np.recarray to be indexed by group ids
    'pvals', sorted np.recarray to be indexed by group ids
    'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
```

Coarse_grain_profiling (without cached files - true cold start)

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
    'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
    'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
ranking genes
  finished: added to `uns['rank_genes_groups']`
    'names', sorted np.recarray to be indexed by group ids
    'scores', sorted np.recarray to be indexed by group ids
    'logfoldchanges', sorted np.recarray to be indexed by group ids
    'pvals', sorted np.recarray to be indexed by group ids
    'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
  Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc6k --out-dir out"
  User time (seconds): 31.09
  System time (seconds): 5.54
  Percent of CPU this job got: 209%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:17.53
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 944928
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 0
  Minor (reclaiming a frame) page faults: 416138
  Voluntary context switches: 34112
  Involuntary context switches: 1135789
  Swaps: 0
  File system inputs: 0
  File system outputs: 198423
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

Use “time.time()” to measure elapsed time for each code section marked with “%%” (instrumented profiling)

```
running Louvain clustering
using the "louvain" package of Traag (2017)
finished: found 7 clusters and added
'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.194s
computing UMAP
finished: added
'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
[TIMING] umap: 6.789s
[TIMING] write results file: 0.276s
ranking genes
finished: added to '.uns['rank_genes_groups']'
'names', sorted np.recarray to be indexed by group ids
'scores', sorted np.recarray to be indexed by group ids
'logfoldchanges', sorted np.recarray to be indexed by group ids
'pvals', sorted np.recarray to be indexed by group ids
'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
Tue Sep 15 22:58:55 2026 rank_genes_groups.prof

80826 function calls (79041 primitive calls) in 2.069 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
24/1   0.000    0.000    2.069    2.069 {built-in method builtins.exec}
1     0.000    0.000    2.069    2.069 <string>:1(<module>)
1     0.000    0.000    2.069    2.069 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/legacy_api_wrap/_init_.py:83(fn_compatible)
1     0.000    0.000    2.069    2.069 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1     0.002    0.002    2.033    2.033 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
8     0.002    0.000    2.004    0.250 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
9     0.005    0.001    1.576    0.175 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
8     0.000    0.000    1.475    0.184 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10012(rank)
8     0.000    0.000    1.475    0.184 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
8     0.000    0.000    1.474    0.184 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
155/154 0.000    0.000    0.186    0.001 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1177(_getitem_)
96     0.000    0.000    0.181    0.002 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1690(_getitem_tuple)
152    0.000    0.000    0.180    0.001 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1720(_getitem_axis)
63     0.000    0.000    0.177    0.003 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1206(_getbool_axis)
56     0.000    0.000    0.176    0.003 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1005(_getitem_tuple_same_dim)

[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 2.176s
```

(iii) Pbmcl0k

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:09)
```

Coarse_grain_profiling (without cached files - true cold start)

```
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:07)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:09)
  Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc10k --out-dir out"
  User time (seconds): 86.29
  System time (seconds): 7.72
  Percent of CPU this job got: 215%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:43.66
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 1892064
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 0
  Minor (reclaiming a frame) page faults: 1455108
  Voluntary context switches: 131516
  Involuntary context switches: 1116942
  Swaps: 0
  File system inputs: 0
  File system outputs: 832587
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

Use “time.time()” to measure elapsed time for each code section marked with “%%” (instrumented profiling)

```
running Louvain clustering
using the "louvain" package of Traag (2017)
finished: found 16 clusters and added
'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.382s
computing UMAP
finished: added
'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
[TIMING] umap: 6.177s
[TIMING] write results file: 0.745s
ranking genes
finished: added to '.uns['rank_genes_groups']'
'names', sorted np.recarray to be indexed by group ids
'scores', sorted np.recarray to be indexed by group ids
'logfoldchanges', sorted np.recarray to be indexed by group ids
'pvals', sorted np.recarray to be indexed by group ids
'pvals adj', sorted np.recarray to be indexed by group ids (0:00:09)
Tue Sep 15 22:57:59 2026 rank_genes_groups.prof

327416 function calls (321615 primitive calls) in 9.154 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>

ncalls  tottime  percall  cumtime  percall  filename:lineno(function)
24/1    0.000    0.000    9.154    9.154 {built-in method builtins.exec}
1       0.000    0.000    9.154    9.154 <string>:1(<module>)
1       0.001    0.001    9.154    9.154 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/legacy api wrap/_init_.py:83(fn_compatible)
1       0.001    0.001    9.154    9.154 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1       0.006    0.006    8.962    8.962 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
17      0.009    0.001    8.892    0.523 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
25      0.016    0.001    6.240    0.250 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
24      0.000    0.000    5.919    0.247 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10012(rank)
24      0.000    0.000    5.919    0.247 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
24      5.914    0.246    5.915    0.246 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
1       0.011    0.011    1.612    1.612 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:174(_basic_stats)
56      0.000    0.000    1.583    0.028 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scipy/sparse/_index.py:29(_getitem_)
32      0.000    0.000    1.536    0.048 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scipy/sparse/_csc.py:140(_get_arrayxslice)
32      0.001    0.000    1.536    0.048 /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scipy/sparse/_compressed.py:651(_minor_index_fancy)
32      1.281    0.040    1.281    0.040 {built-in method scipy.sparse._sparsetools.csr_column_index2}

[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 9.168s
```

(b) Create the slurm batch scripts for each experiment (including pbmc3k).

SLURM Batch Scripts

1) pbmc3k

```
#!/bin/bash
#SBATCH --job-name=scanpy_pbmc3k
#SBATCH --output=logs/pbmc3k_%j.out
#SBATCH --error=logs/pbmc3k_%j.err
#SBATCH --time=00:30:00
#SBATCH --partition=<partition>    # use available partitions with: sinfo
#SBATCH --account=<account>        # redacted for privacy
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=8G

#environment
source ~/miniconda3/etc/profile.d/conda.sh
conda activate bmi500

# /usr/bin/time -v gives coarse-grain stats (wall time, CPU time, peak memory)
# the script itself prints [TIMING] lines per ### section (instrumented)
# and runs cProfile on rank_genes_groups (fine-grain), saved to rank_genes_groups.prof
/usr/bin/time -v python scanpy_pbmc_time.py \
    --data-dir data \
    --data-set pbmc3k \
    --out-dir out

# keep this dataset's fine-grain profile separate from the others
mv rank_genes_groups.prof pbmc3k_rank_genes_groups.prof
```

2) pbmc_6k

```
#!/bin/bash
#SBATCH --job-name=scanpy_pbmc6k
#SBATCH --output=logs/pbmc6k_%j.out
#SBATCH --error=logs/pbmc6k_%j.err
#SBATCH --time=00:30:00
#SBATCH --partition=<partition>    # use available partitions with: sinfo
#SBATCH --account=<account>        # redacted for privacy
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=8G

#environment
source ~/miniconda3/etc/profile.d/conda.sh
```

```
conda activate bmi500
```

```
# /usr/bin/time -v gives coarse-grain stats (wall time, CPU time, peak memory)
# the script itself prints [TIMING] lines per ### section (instrumented)
# and runs cProfile on rank_genes_groups (fine-grain), saved to rank_genes_groups.prof
/usr/bin/time -v python scanpy_pbmc_time.py \
    --data-dir data \
    --data-set pbmc6k \
    --out-dir out
```

```
# keep this dataset's fine-grain profile separate from the others
mv rank_genes_groups.prof pbmc6k_rank_genes_groups.prof
```

3) pbmc_10k

```
#!/bin/bash
#SBATCH --job-name=scanpy_pbmc10k
#SBATCH --output=logs/pbmc10k_%j.out
#SBATCH --error=logs/pbmc10k_%j.err
#SBATCH --time=00:30:00
#SBATCH --partition=<partition>    # use available partitions with: sinfo
#SBATCH --account=<account>        # redacted for privacy
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=8G
```

```
#environment
source ~/miniconda3/etc/profile.d/conda.sh
conda activate bmi500
```

```
# /usr/bin/time -v gives coarse-grain stats (wall time, CPU time, peak memory)
# the script itself prints [TIMING] lines per ### section (instrumented)
# and runs cProfile on rank_genes_groups (fine-grain), saved to rank_genes_groups.prof
/usr/bin/time -v python scanpy_pbmc_time.py \
    --data-dir data \
    --data-set pbmc10k \
    --out-dir out
```

```
# keep this dataset's fine-grain profile separate from the others
mv rank_genes_groups.prof pbmc10k_rank_genes_groups.prof
```

(c) Console outputs for all dataset runs (pbmc3k, pbmc6k, pbmc10k)

(i) /usr/bin/time -v python scanpy_pbmc_time.py --data-dir data --data-set pbmc3k --out-dir out

```
[TIMING] timing setup: 0.000s
[TIMING] imports: 0.965s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6louvain==0.8.2 pynndescent==0.6.0
using 1 threads
[TIMING] scanpy settings: 3.569s
[TIMING] argument parsing: 0.000s
... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
[TIMING] I/O: read 10x data: 0.076s
filtered out 19024 genes that are detected in less than 3 cells
[TIMING] preprocessing: basic filtering: 0.034s
normalizing counts per cell
  finished (0:00:00)
[TIMING] QC metrics + normalization: 0.024s
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
[TIMING] highly variable genes: 0.033s
... as 'zero_center=True', sparse input is densified and may lead to large memory consumption
[TIMING] scale: 0.031s
[TIMING] report adata (no-op): 0.000s
computing PCA
  with n_comps=30
  finished (0:00:00)
[TIMING] PCA: 0.296s
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to '.uns['neighbors']'
  '.obspl['distances']', distances for each pair of neighbors
  '.obspl['connectivities']', weighted adjacency matrix (0:00:02)
[TIMING] neighbors: 2.384s
[TIMING] paga fix (no-op): 0.000s
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.064s
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:03)
[TIMING] umap: 3.422s
[TIMING] write results file: 0.144s
ranking genes
  finished: added to '.uns['rank_genes_groups']'
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:00)
Wed Sep 16 13:47:41 2026   rank_genes_groups.prof

42485 function calls (41332 primitive calls) in 0.972 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>
```

ncalls	tottime	percall	cumtime	percall	filename:lineno(function)
24/1	0.000	0.000	0.972	0.972	{built-in method builtins.exec}
1	0.000	0.000	0.972	0.972	<string>:1(<module>)
1	0.000	0.000	0.972	0.972	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/legacy_api_wrap/_init_.py:83(fn_compatible)
1	0.000	0.000	0.972	0.972	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1	0.001	0.001	0.950	0.950	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
5	0.002	0.000	0.931	0.186	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
5	0.002	0.000	0.751	0.150	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
4	0.000	0.000	0.696	0.174	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10012(rank)
4	0.000	0.000	0.696	0.174	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
4	0.695	0.174	0.696	0.174	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
60/59	0.000	0.000	0.077	0.001	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1177(_getitem_)
41	0.000	0.000	0.075	0.002	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1690(_getitem_tuple)
57	0.000	0.000	0.074	0.001	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1720(_getitem_axis)
23	0.000	0.000	0.073	0.003	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1206(_getbool_axis)
23	0.000	0.000	0.072	0.003	/home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:4164(_take_with_is_copy)

```
[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 1.072s
Command being timed: "python -W ignore scanpy_pbmc_time.py --data-dir data --data-set pbmc3k --out-dir out"
User time (seconds): 23.39
System time (seconds): 4.90
Percent of CPU this job got: 227%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:12.43
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 870488
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 0
Minor (reclaiming a frame) page faults: 207034
Voluntary context switches: 15995
Involuntary context switches: 1155135
Swaps: 0
File system inputs: 0
File system outputs: 87464
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

(ii) /usr/bin/time -v python scanpy_pbmc_time.py --data-dir data --data-set pbmc6k --out-dir out

```
[TIMING] timing setup: 0.000s
[TIMING] imports: 0.947s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
[TIMING] scanpy settings: 3.576s
[TIMING] argument parsing: 0.000s
... reading from cache file cache/data-pbmc6k-filtered_gene_bc_matrices-matrix.h5ad
[TIMING] I/O: read 10x data: 0.115s
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
[TIMING] preprocessing: basic filtering: 0.057s
normalizing counts per cell
  finished (0:00:00)
[TIMING] QC metrics + normalization: 0.039s
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions norm', float vector (adata.var)
[TIMING] highly variable genes: 0.049s
... as 'zero_center=True', sparse input is densified and may lead to large memory consumption
[TIMING] scale: 0.059s
[TIMING] report adata (no-op): 0.000s
computing PCA
  with n_comps=30
  finished (0:00:00)
[TIMING] PCA: 0.374s
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to '.uns['neighbors']'
  '.obsp['distances']', distances for each pair of neighbors
  '.obsp['connectivities']', weighted adjacency matrix (0:00:02)
[TIMING] neighbors: 2.392s
[TIMING] paga fix (no-op): 0.000s
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.185s
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
[TIMING] umap: 0.557s
[TIMING] write results file: 0.257s
ranking genes
  finished: added to '.uns['rank_genes_groups']'
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals adj', sorted np.recarray to be indexed by group ids (0:00:01)
Wed Sep 16 13:52:10 2026   rank_genes_groups.prof

      88826 function calls (79041 primitive calls) in 1.931 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>
```

```
ncalls  tottime  percall  cumtime  percall  filename:lineno(function)
24/1    0.000    0.000    1.931    1.931  {built-in method builtins.exec}
1       0.000    0.000    1.931    1.931  <string>:1(<module>)
1       0.000    0.000    1.931    1.931  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/legacy_api_wrap/_init_.py:83(fn_compatible)
1       0.000    0.000    1.931    1.931  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1       0.002    0.002    1.893    1.893  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
0       0.002    0.000    1.864    0.223  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
0       0.005    0.001    1.449    0.161  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
0       0.000    0.000    1.346    0.168  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:1012(rank)
0       0.000    0.000    1.346    0.168  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
0       1.345    0.168    1.345    0.168  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
155/154 0.000    0.000    0.185    0.001  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1177(_getitem_)
96       0.000    0.000    0.180    0.002  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1690(_getitem_tuple)
152      0.000    0.000    0.179    0.001  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1720(_getitem_axis)
63       0.000    0.000    0.176    0.003  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1286(_getitem_axis)
56       0.000    0.000    0.175    0.003  /home/pks/miniconda3/envs/bmi500/lib/python3.11/site-packages/pandas/core/indexing.py:1005(_getitem_tuple_same_dim)

[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 2.032s
Command being timed: "python -W ignore scanpy_pbmc_time.py --data-dir data --data-set pbmc6k --out-dir out"
User time (seconds): 29.91
System time (seconds): 5.10
Percent of CPU this job got: 206%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:16.96
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 934008
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 1
Minor (reclaiming a frame) page faults: 301012
Voluntary context switches: 27211
Involuntary context switches: 441375
Swaps: 0
File system inputs: 208
File system outputs: 161393
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

(iii) /usr/bin/time -v python scanpy_pbmc_time.py --data-dir data --data-set pbmc10k --out-dir out

```
[TIMING] timing setup: 0.000s
[TIMING] imports: 0.965s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.8.2 pyndescent==0.6.0
using 1 threads
[TIMING] scanpy settings: 3.666s
[TIMING] argument parsing: 0.000s
... reading from cache file cache/data-pbmc10k-filtered_gene_bc_matrices-matrix.h5ad
[TIMING] I/O: read 10x data: 0.439s
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
[TIMING] preprocessing: basic filtering: 0.280s
normalizing counts per cell
  finished (0:00:00)
[TIMING] QC metrics + normalization: 0.282s
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions norm', float vector (adata.var)
[TIMING] highly variable genes: 0.254s
... as 'zero center=True', sparse input is densified and may lead to large memory consumption
[TIMING] scale: 0.143s
[TIMING] report adata (no-op): 0.000s
computing PCA
  with n_comps=30
  finished (0:00:00)
[TIMING] PCA: 0.667s
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to '.uns['neighbors']'
  '.obs['distances']', distances for each pair of neighbors
  '.obs['connectivities']', weighted adjacency matrix (0:00:16)
[TIMING] neighbors: 16.728s
[TIMING] paga fix (no-op): 0.000s
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
[TIMING] clustering: louvain: 0.379s
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:06)
[TIMING] umap: 6.248s
[TIMING] write results file: 0.730s
ranking genes
  finished: added to '.uns['rank_genes_groups']'
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals adj', sorted np.recarray to be indexed by group ids (0:00:00)
Wed Sep 16 13:53:56 2026   rank_genes_groups.prof

      327416 function calls (321615 primitive calls) in 8.851 seconds

Ordered by: cumulative time
List reduced from 1063 to 15 due to restriction <15>
```

```
ncalls  tottime  percall  cumtime  percall  filename:lineno(function)
24/1    0.000    0.000    8.851    8.851  {built-in method builtins.exec}
1       0.000    0.000    8.851    8.851  <string>:1(<module>)
1       0.001    0.001    8.851    8.851  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/legacy_api_wrap/_init_.py:83(fn_compatible)
1       0.001    0.001    8.850    8.850  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:469(rank_genes_groups)
1       0.006    0.006    8.659    8.659  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:398(compute_statistics)
17      0.009    0.001    8.587    0.505  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:277(wilcoxon)
25      0.016    0.001    8.590    0.238  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:45(_ranks)
24      0.000    0.000    5.627    0.234  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/pandas/core/generic.py:10012(rank)
24      0.000    0.000    5.627    0.234  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/pandas/core/generic.py:10127(ranker)
24      5.623    0.234    5.624    0.234  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/pandas/core/algorithms.py:1067(rank)
1       0.012    0.012    1.604    1.604  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scanpy/tools/_rank_genes_groups.py:174(_basic_stats)
56      0.000    0.000    1.575    0.028  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scipy/sparse/_index.py:29(_getitem_)
32      0.000    0.000    1.528    0.048  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scipy/sparse/_csc.py:140(_get_arrayxslice)
32      0.001    0.000    1.528    0.048  /home/pks/miniconda3/envs/bm1500/lib/python3.11/site-packages/scipy/sparse/_compressed.py:651(_minor_index_fancy)
32      1.276    0.040    1.276    0.040  {built-in method scipy.sparse._sparsetools.csr_column_index2}

[TIMING] rank_genes_groups (wilcoxon, cProfile'd): 8.864s
Command being timed: "python -W ignore scanpy_pbmc_time.py --data-dir data --data-set pbmc10k --out-dir out"
User time (seconds): 60.83
System time (seconds): 7.60
Percent of CPU this job got: 170%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:40.19
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1343672
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 4
Minor (reclaiming a frame) page faults: 1206018
Voluntary context switches: 92030
Involuntary context switches: 547498
Swaps: 0
File system inputs: 1776
File system outputs: 590614
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

Coarse-grain Summary

Metric	pbmc3k	pbmc6k	pbmc10k
Elapsed (wall clock) time	12.43 s	16.96 s	40.19 s
User (CPU) time	23.39 s	29.91 s	60.83 s
System time	4.90 s	5.10 s	7.68 s
CPU utilization	227%	206%	170%
Max resident memory	850 MB	912 MB	1312 MB
Minor page faults	207,034	301,012	1,206,018

Highlight:

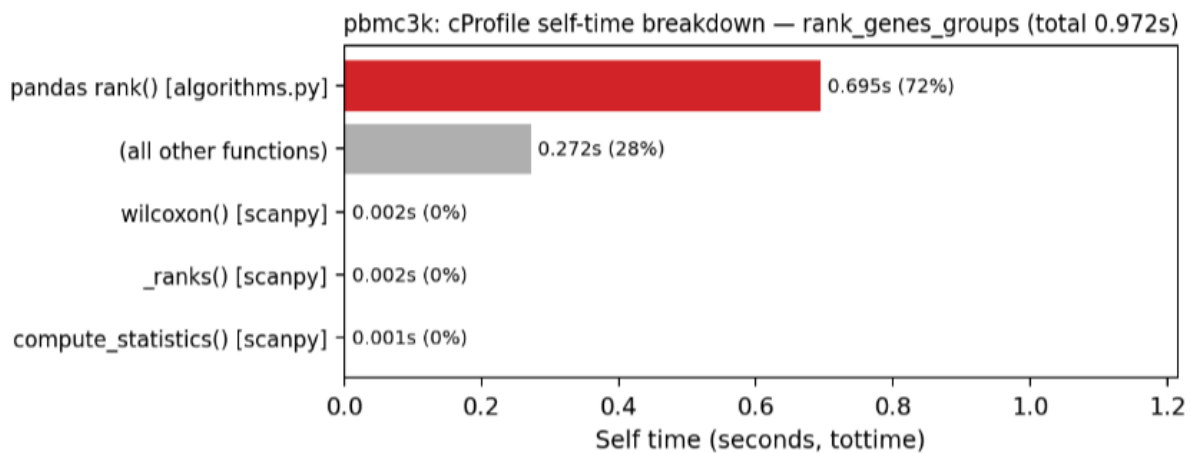
- 1) Wall clock time scales worse than linearly between pbmc6k and pbmc10k.
- 2) CPU utilization drops from pbmc3k->pbmc6k->pbmc10k. This combined with above indicates the pbmc10k run spends more time in a less-parallelized stage.
- 3) Peak memory grows with dataset size consistent with dense-matrix operations.

Instrumented Timing (bottleneck by section)

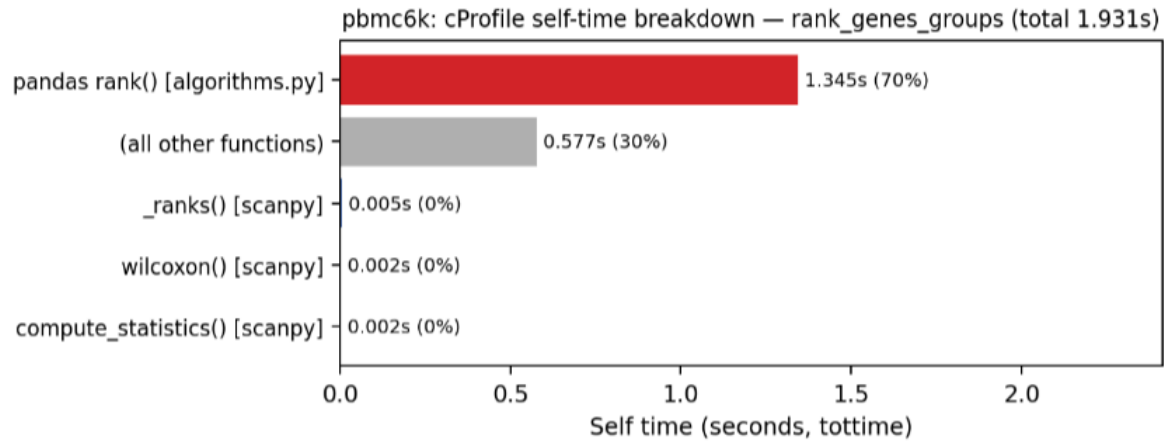
Dataset	Largest Section	Time	% of total runtime
pbmc3k	umap	3.42 s	28%
pbmc6k	umap	6.56 s	39%
pbmc10k	neighbors	16.73 s	42%

Major bottleneck: The dominant section shifts with scale. At pbmc3k and pbmc6k, UMAP is the single largest section. At pbmc10k, neighbors (KNN graph construction) is largest.

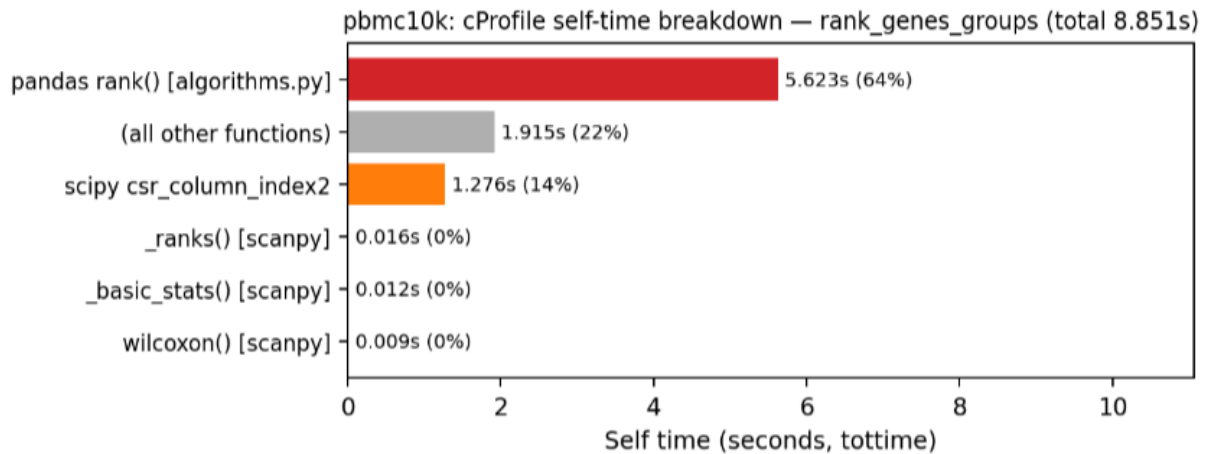
Fine-grain cProfile visualization (rank_genes_group)



The Wilcoxon rank-sum test's ranking step is the core cost even at the smallest scale (pandas' rank)

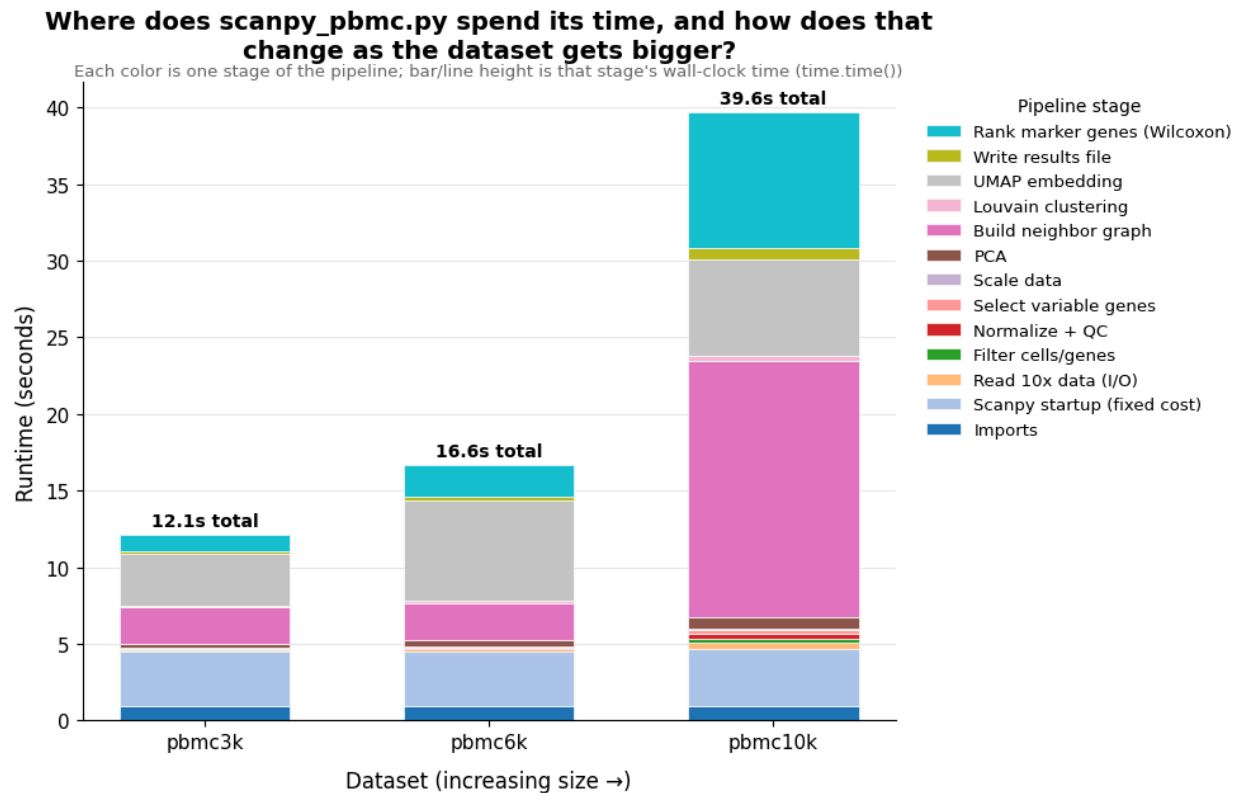


Same pattern: pandas' rank dominates which is consistent with previous findings.



Same pattern: pandas' rank dominates but now we can see other components growing as well.

(d) Plot the instrumented profile results for each code section vs dataset size (dataset on x-axis, run time on y-axis). Use stacked bar, line, or stacked area plot to show runtime growth dataset size.



(e) Where are the performance bottlenecks? If the data size were to grow to 20K, what are the expected running times for the code sections?

Across all the datasets, we can identify the following bottlenecks:

(i) Marker gene ranking (Wilcoxon): From the cProfile data, pandas' generic rank() function consumes a lot of time at every scale. It calls the method per group per gene rather than being optimized for repeated group-wise ranking.

(ii) Neighbor graph construction: With increasing data size, it scales non-linearly as evident in pbmc10k timings. So this is marked as the biggest unknown when scaling to even greater data.

At 20k data, we can extrapolate the following:

Neighbor graph would take anywhere from 25-30 seconds

Rank marker genes: 18-22 seconds

UMAP: around 6 seconds

Rest of the stages (each): around 1-1.5 seconds

Total (sum of stages): Somewhere around 60-80 seconds

Link to GitHub Repository for everything else: https://github.com/pks716/bmi500_hw3

Code review text files are also available in the same repository.